

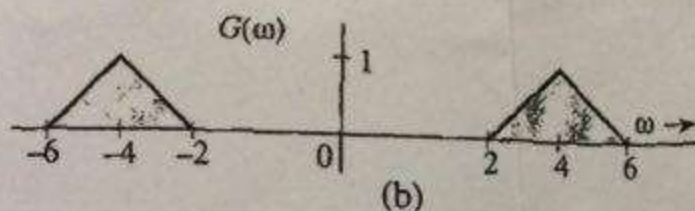


Program: BS(EE)
Duration: 60 Minutes
Paper Date: 10-11-16
Section: ALL
Exam: Midterm-II

Course Code: EE223
Semester: Fall 2016
Total Marks: 50
Weight: 20%
Page(s): 1
Roll No:

Instruction/Notes: Closed book, closed notes, Calculator allowed

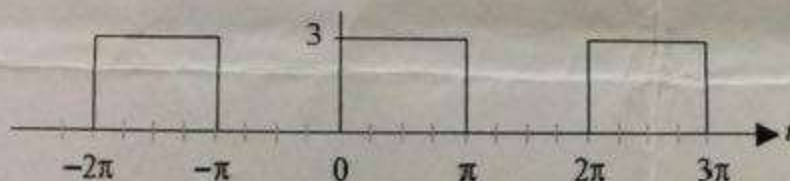
Q.1 (15): $F(\omega) = \delta(\omega - 4) + \delta(\omega + 4)$ and $G(\omega)$ is plotted below;



Plot $H(\omega) = F(\omega) * G(\omega)$

Q.2 (20):

a) Compute the exponential Fourier series of the following function $x(t)$.



b) Find the Fourier coefficients ($a_i, b_i, i=0,1,\dots$) of the following Fourier Series;

$$x(t) = 3 + \cos\left(4t + \frac{\pi}{4}\right) + \sin\left(10t + \frac{\pi}{3}\right).$$

Q.3 (15): The following causal systems are described by input $[x(t)]$ and output $[y(t)]$ equations. Determine if the systems are 1) linear/ Non-linear, 2) time-invariant / time-varying, 3) memory-less / with memory? For each part, write your choices clearly in a box.

a) $y(t) = x(t - 2)$

b) $y(t) = U(x(t))$

c) $y(t) = tx(t)$